ĐẠI HỌC HOA SEN

TÔN TRỌNG SỰ KHÁC BIỆT

WEEK 7: TO-BE PROCESS DESIGN & REQUIREMENTS REFINEMENT

Course: Business Systems Analysis

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Faculty: Faculty of Technology

Learning Focus

By the end of this session, you should be able to:

- Five reviewed business process descriptions.
- A process relationship map.
- Evidence-based business problems and information needs.
- Draft TO-BE process models for the assigned processes.
- Refined Business Requirements (BRs) and Functional Requirements (FRs).
- A requirements traceability matrix and a list of unresolved questions.

Do not simply add more requirements. Make your existing requirements more defensible.

How will we work today?

Time	Activity	Main Output
10–28 min	Activity 1 – Individual Process Deep Dive	Five reviewed process descriptions
28–40 min	Activity 2 – Integrate the Five Processes	Process relationship map
40–55 min	Activity 3 – Evidence-Based Problem Analysis	Problems and information needs
55–75 min	Activity 4 – Draft the TO-BE Process	Proposed process improvements
75–95 min	Activity 5 – Refine BRs and FRs	Revised requirements
95–110 min	Activity 6 – Requirements Traceability	Updated traceability matrix
110–120 min	Activity 7 – Internal Peer Review	Review notes
120–130 min	Activity 8 – Cross-Group Peer Review	External feedback
130–135 min	Exit Ticket	Individual reflection

Activity 1 — Individual Process Deep Dive

Each member is responsible for reviewing one assigned business process.

Task: Review the existing AS-IS model and interview evidence. Complete the process description below.

Element	Questions to Answer
Process Name	What business activity does this process perform?
Trigger	What event starts the process?
Start Point	Where does the process begin?
Main Activities	What actually happens, and in what sequence?
Main Actors	Which roles or departments participate?
Information Used/Produced	What information is needed, created, or updated?
Handoffs	When and how is work transferred between actors?
Decisions & Exceptions	What decisions or exceptional situations occur?
End Point	What final state indicates completion?
Business Outcome	What business result does the process achieve?
Evidence Sources	Which interview notes or observations support the description?
Uncertain Information	What still needs to be confirmed?

Use a swimlane / cross-functional flowchart or BPMN notation consistent with the modeling approach taught in class.

Activity 2 — Integrate the Five Business Processes

Objective: Understand how the five processes relate to the overall operation of the enterprise.

Team task

- Place the five assigned processes on one process relationship map.
- Identify which process may occur before or after another process.
- Identify the information, document, product, or status transferred between processes.
- Highlight dependencies and possible gaps between processes.
- Mark any relationship that has not yet been verified.

Process Relationship Map

Upstream Process	Output / Information Transferred	Downstream Process	Evidence / Status
Process A	What is received or transferred?	Process B	Verified / Uncertain
Process B	What is received or transferred?	Process C	Verified / Uncertain

Discussion questions

- Does one process depend on the output of another?
- Is information transferred consistently between processes?
- Are there handoffs that may cause delays or errors?
- Are the process boundaries clear?

Activity 3 — Evidence-Based Problem Analysis

Objective: Identify business problems from the team's verified AS-IS analysis.

- Do not start by suggesting a technology feature. First determine what is wrong with the current operation and why it matters.

Task

- For each important issue, complete the following table.

Evidence Source	AS-IS Observation	Business Problem	Affected Stakeholder	Information Need	Business Impact	Open Question
Interview / record / observation	What happens now?	What is the actual difficulty?	Who is affected?	What information is needed?	Why does it matter?	What remains uncertain?

Activity 3 — Evidence-Based Problem Analysis

- Example

Element	Example
Evidence source	Interview with Sales Staff
AS-IS observation	Sales Staff contact the Warehouse to confirm stock availability.
Business problem	Sales Staff cannot directly access reliable stock availability information when preparing an order.
Affected stakeholder	Sales Staff
Information need	Product availability information before order confirmation
Business impact	Potential delays in order processing
Open question	How frequently does the information delay occur?

This is an illustrative example. Each team must analyze its own evidence.

Deliverable: A prioritized list of business problems and associated information needs.

Activity 4 — Draft the TO-BE Process (20 minutes)

Objective: Propose a future process that addresses verified business problems.

A TO-BE process describes how the business process is proposed to operate in the future. It is not yet a detailed system design.

Task

- Select a core business process and review its verified AS-IS model.
- Identify the activities, decisions, handoffs, or information exchanges that need to change.
- Develop a TO-BE swimlane / cross-functional flowchart or BPMN model showing the proposed future process.
- Clearly identify the actors responsible for activities and the sequence of work.
- Explain how the proposed changes address the verified business problems.
- Record assumptions and unresolved questions.

Activity 4 — Draft the TO-BE Process (20 minutes)

AS-IS vs. TO-BE Analysis

AS-IS: Current Process	TO-BE: Proposed Process
What currently happens?	What should change?
Where does the problem occur?	Where will the process improve?
What information is missing or delayed?	What information should be available, and when?
Which handoffs create difficulty?	How should the handoffs change?

Example

- AS-IS: Sales Staff contact the Warehouse to obtain stock availability information.
- TO-BE proposal: Sales Staff receive reliable stock availability information before confirming an order.
- Expected improvement: Fewer manual inquiries and potentially shorter order-processing time.

Review questions

- Does the proposed process directly address a verified problem?
- Are the actors, activities, decisions, and handoffs understandable?
- Is the proposed change realistic from a business perspective?
- Have assumptions been clearly identified?

Activity 5 — Refine Business and Functional Requirements (20 minutes)

Objective: Improve the team's Week 6 BR/FR drafts using the deeper process analysis and proposed TO-BE changes.

Do not rewrite definitions from Week 6. Apply the concepts to your team's actual project.

Task

- Review each existing requirement and decide whether to Keep, Revise, Remove, or Investigate it.

Requirement ID	Existing Requirement	Linked Problem / Need	Review Decision	Revised Requirement	Open Question
BR-01 / FR-01	Existing draft	Problem or information need	Keep / Revise / Remove / Investigate	Revised statement	Unresolved issue

Activity 5 — Refine Business and Functional Requirements (20 minutes)

Requirements review criteria

- Is the requirement justified by verified evidence or an explicitly stated business need?
- Is the BR about a business outcome or capability?
- Is the FR specific enough to describe observable system behavior?
- Can the FR be tested or verified?
- Is the requirement consistent with the proposed TO-BE process?
- Does it introduce a solution assumption that has not been justified?

Example

Business problem: Sales Staff lack timely access to reliable stock availability information.

- BR-01: The business needs order confirmation to be based on reliable product availability information.
- FR-01: The system shall display available product quantities to authorized Sales Staff during order preparation.
- Important: Avoid vague requirements such as “The system should be faster” or “The system should improve business operations.” State the required outcome or observable behavior.
- Deliverable: A revised BR/FR list with IDs and review decisions.

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Activity 6 — Requirements Traceability (15 minutes)

Objective: Verify that each important requirement can be traced back to an identified business need.

Task

- Select your team's most important requirements and complete the traceability matrix.
- Required reasoning chain

  Evidence → AS-IS Observation → Business Problem → Information Need → BR → FR

Example

Evidence / Source	AS-IS Observation	Problem / Need	BR ID	FR ID	Related Process
Sales interview	Sales contacts Warehouse to confirm stock.	Sales needs reliable stock information before confirming an order.	BR-01	FR-01	Processing Customer Orders

Activity 6 — Requirements Traceability (15 minutes)

Validation questions

- Can the team explain why each requirement is needed?
- Does the requirement address the identified problem or information need?
- Is the FR consistent with the BR?
- Does the requirement support the proposed TO-BE process?
- If the evidence does not support a requirement, should it be revised or investigated?

Important: Do not create artificial links merely to complete the table. Mark missing evidence or unclear relationships for follow-up.

Deliverable: An updated traceability matrix and a list of requirements needing further investigation.

Activity 7 & 8 — Peer Review

Internal Peer Review — 10 minutes

Each member reviews another member's process and requirements.

Checklist

- Is the process scope clear?
- Are the trigger, activities, actors, handoffs, and end point understandable?
- Are important problems supported by evidence?
- Does the TO-BE proposal address an identified problem?
- Are the BRs and FRs clear and justified?
- Can the requirements be traced to the business need?

Each reviewer records:

- One strength.
- One issue requiring clarification or revision.
- One question the process owner should answer.

Cross-Group Peer Review — 10 minutes

Exchange one selected process model and its related TO-BE proposal or BR/FR table with another group.

Review the submitted work without assuming facts that are not documented.

Provide feedback:

Review Item	Feedback
One strong point	What is clear, logical, or well-supported?
One issue	What is unclear, unsupported, or inconsistent?
One question	What evidence or explanation is still needed?

Review principle: Give specific feedback that helps improve the analysis, not simply “good” or “needs improvement.”

Week 7 Deliverables & Next Steps

Team Deliverables

Before leaving class, ensure that your team has:

- ❑ Five reviewed business process descriptions.
- ❑ One process relationship map.
- ❑ An evidence-based problem and information-needs analysis.
- ❑ Draft TO-BE models for the assigned processes, prioritizing the core processes.
- ❑ A refined BR/FR list with requirement IDs.
- ❑ An updated requirements traceability matrix.
- ❑ Internal and cross-group peer review notes.
- ❑ A list of assumptions and unresolved questions.

Individual Exit Ticket — 5 minutes

Answer these questions individually.

- Which business problem did your process analysis clarify most clearly?
- How does your proposed TO-BE process address that problem?
- Which requirement did you revise after reviewing the evidence? Why?
- What is one unresolved question your team must address before finalizing Phase 1?

Next Week — Week 8: Phase 1 Report Review & Presentation Preparation

- Review consistency across enterprise context, stakeholders, AS-IS, problems, TO-BE, BRs, and FRs.
- Resolve contradictions and correct unsupported claims.
- Complete peer review and revise the report.
- Prepare the Week 9 presentation and allocate speaking responsibilities.